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Biased Auctioneers

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01 / MOTIVATION

The Research Question



"Are human pre-sale estimates informationally efficient, or systematically skewed by cognitive and strategic biases?"

Testing expert valuation accuracy using machine learning benchmarks.

02 / THEORETICAL FRICTION

Why Estimates Deviate

Asset Heterogeneity

Unique items trading very infrequently.

Cognitive Biases

Slow adjustment to recent trends.

Strategic Incentives

Luring sellers vs. attracting bidders.

03 / DATA (2008–2015)

Global Auction Dataset

1.2M
Painting auction lots

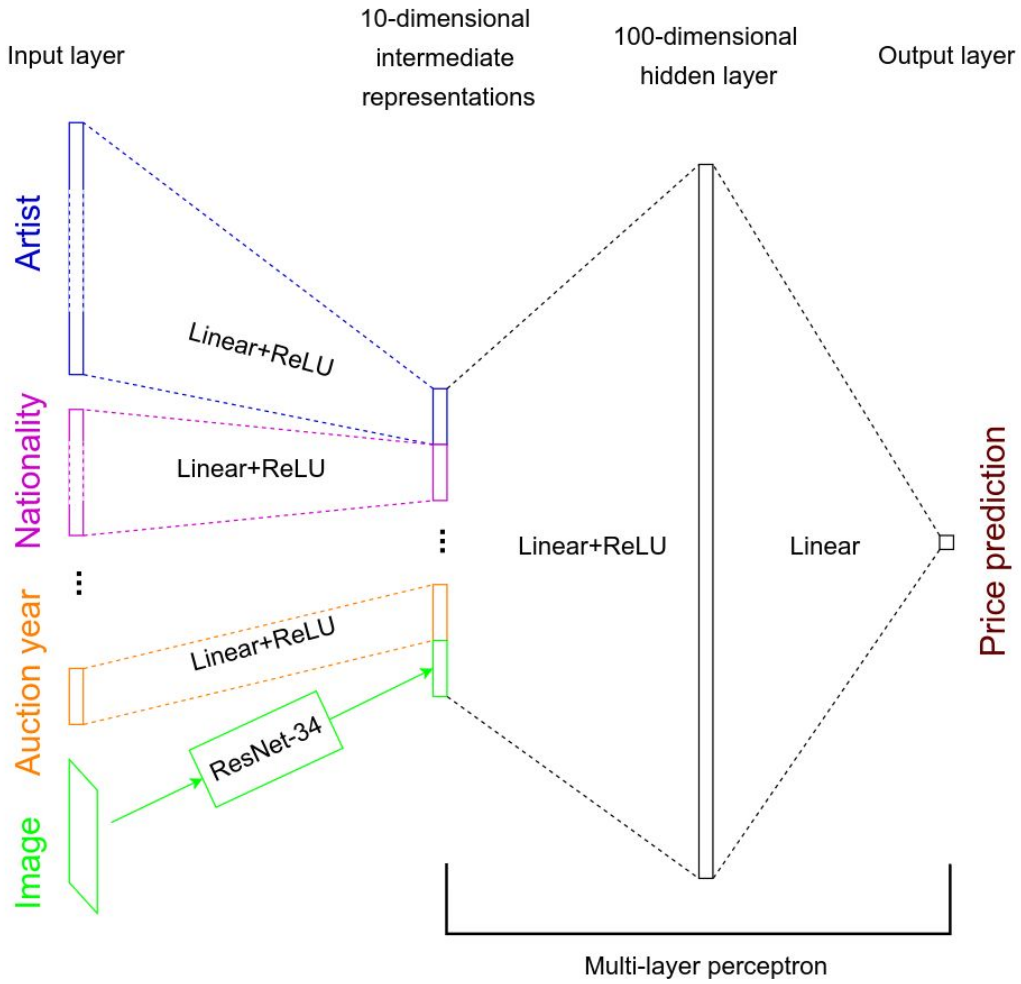
130K
Individual artists tracked

$61.2K
Mean price ($3.3K median)

35%
Unsold "buy-in" rate

04 / ARCHITECTURE

Neural Network Valuation



Out-of-Sample Testing

2015 counterfactual price (ML)

05 / CLASS LINK

Hedonic OLS vs. Neural Networks

Linear Hedonic Model

Additive Form: Misses complex variable interactions.

Text-Only: Blind to visual composition.

Test Fit: Explains 67.7% price variance.

Deep Neural Network

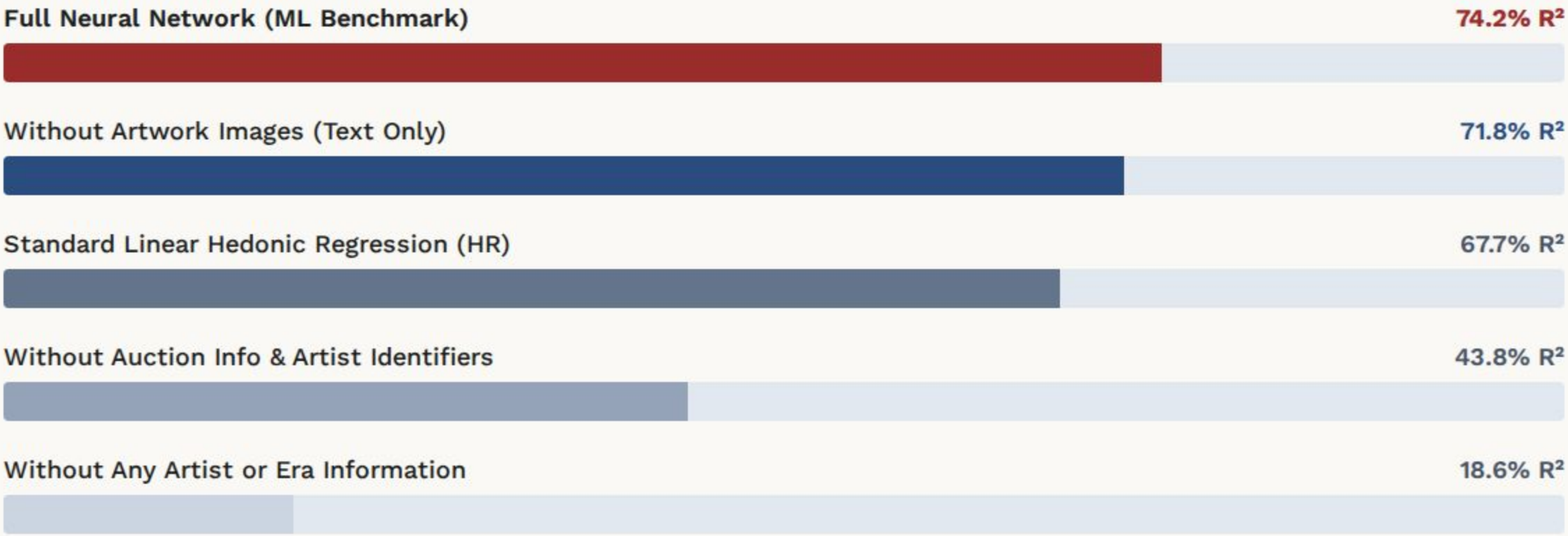
Non-Linear: Captures artist-specific trait interactions.

Regularized: 20% input dropout prevents overfitting.

Test Fit: Explains 74.2% price variance.

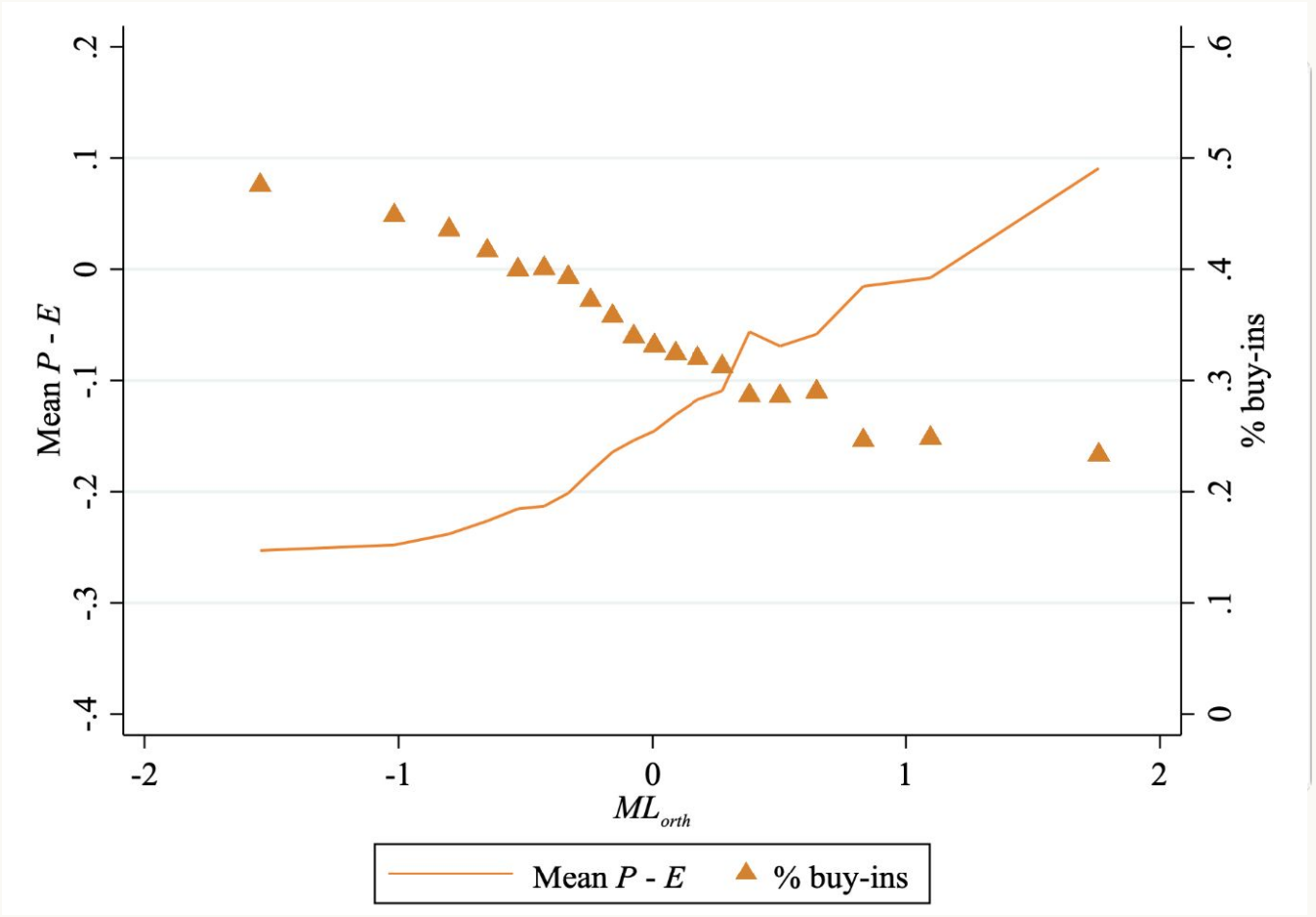
06 / TABLE II RESULTS

What Drives Art Price Predictability?



07 / REQUIRED EXHIBIT (FIGURE 10)

Informational Inefficiency of Estimates



High ML vs. Estimate: Prices beat estimates; buy-ins drop to 25%.

Low ML vs. Estimate: Reserve prices set too high; 45%+ fail to sell.

Core Takeaway: Human estimates omit public information captured by ML.

Predicting the Auctioneer's Error

Artist Persistence

Past artist valuation errors repeat out-of-sample.

House Persistence

Specific auction houses stay systematically aggressive.

Ex-Ante Predictability

ML predicts errors without seeing the estimate.

09 / CRITIQUE

Are the Results Convincing?

Key Strengths

Massive Scale: Near-totality of global auctions.

Clean Split: Strict 2015 out-of-sample testing.

Where ML Wins: Low-price, low-dispersion artists.

Limitations & Caveats

Unobserved Factors: Blind to provenance and condition.

Imputed Buy-Ins: Unsold works fixed at 75% low estimate.

Mechanism Ambiguity: Cannot separate cognitive vs. strategic.

Broader Market Implications

Investor Harm

Credulous buyers overpay on inflated estimates.

Real Estate & iBuyers

Auditing human appraisers in illiquid markets.

Market Transparency

AI benchmarks check intermediary market power.



11 / CLASS DISCUSSION (10 MINUTES)

Questions for Discussion

Question 01 • Market Equilibrium

If buyers and sellers adopt open AI valuation tools, how will auction houses adapt their strategic use of pre-sale estimates?

Question 02 • Policy & Ethics

Should intermediaries in illiquid markets (art, housing) be required to disclose automated benchmark valuations alongside human appraisals?